Report - Version 13: Evaluation and Controlled Generation

Introduction

Version 13 keeps the Version 12 WikiText-103 Raw v1 dataset, BPE tokenizer, two-block decoder-only Transformer, optimization settings, and checkpoint pipeline unchanged. The new concept is held-out evaluation together with explicit generation controls.

The validation-selected BEST checkpoint is restored before the test split is measured. Cross-entropy is complemented by perplexity, top-1 accuracy, and top-5 accuracy. Temperature, top-k, and top-p are implemented directly from final-position logits, so decoding changes can be studied without changing model parameters.

Version 13 base model restored from BEST

```
-> cross-entropy + perplexity + top-1/top-5 accuracy
-> controlled next-token distributions
-> greedy / temperature / top-k / top-p generation
-> attention inspection + tests
```

Goal of Version 13

- preserve the complete Version 12 training and checkpoint experiment
- evaluate only the validation-selected model on the held-out test split
- report cross-entropy, perplexity, top-1 accuracy, and top-5 accuracy
- implement temperature, top-k, and top-p directly from logits
- compare decoding rules with the same prompt, 40 generated BPE tokens, and seed 42
- keep the implementation explicit and reproducible

Architecture and continuity

```
WikiText-103 Raw v1
 -> selected rows 10,000 / 1,000 / 1,000
 -> BPE vocabulary 2,048
 -> used sequences 12k / 2k / 2k, context 16
 -> token + positional embeddings
 -> Transformer block 1
 -> Transformer block 2
 -> vocabulary logits
 -> train 1-10 -> restore LAST -> resume 11-20
 -> restore BEST -> held-out evaluation + controlled generation
```

Continuity with Version 12

Tokenizer learning still uses only selected WikiText training text; validation and test do not influence vocabulary construction. Batch size 32, learning rate 1.0, the 20-epoch budget, epoch-specific deterministic shuffling, and the minimal LAST/BEST checkpoint semantics are preserved.

New post-training stage

- test data is evaluated only after BEST is restored
- all evaluation metrics are derived from the same next-token logits
- decoding controls transform logits/probabilities only and never update weights
- all generation modes share the same held-out prompt and generation length
- stochastic sampling uses a local NumPy generator with seed 42

Trainable parameters

Component	Parameters
Token embeddings	2048 x 8 = 16384
Positional embeddings	16 x 8 = 128
Transformer block 1	800
Transformer block 2	800
Output weights	8 x 2048 = 16384
Total	34496

Each Transformer block still contributes exactly 800 parameters:

Part	Shape	Params
Query projection	8 x 16	128
Key projection	8 x 16	128
Value projection	8 x 16	128
Attention output	16 x 8	128
LayerNorm 1 scale + shift	8 + 8	16
Feed-forward input	8 x 16	128
Feed-forward output	16 x 8	128
LayerNorm 2 scale + shift	8 + 8	16
Per block		800

Checkpoint state

State	Contents	Purpose
LAST	model + completed_epoch + best metadata	resume
BEST	model + best epoch/loss	selection
FINAL	fresh model restored from BEST	test + analysis

Version 13 does not add trainable parameters. The complete model remains at 23 trainable TensorFlow variables and exactly 34,496 parameters.

Held-out evaluation

Metrics are computed token-by-token on the 2,000 selected test sequences after BEST restoration. With context length 16, the held-out evaluation covers 32,000 target-token positions.

Metric	Definition
Cross-entropy	average negative log-probability of the true token
Perplexity	exp(cross-entropy)
Top-1 accuracy	true token equals the highest-logit token
Top-5 accuracy	true token appears among the five highest logits
Evaluated tokens	2,000 x 16 = 32,000

Reference baseline

The uniform-vocabulary reference remains log(2048) for cross-entropy and 2048 for perplexity. These metrics describe token prediction; they do not directly measure factuality, coherence, or usefulness.

Controlled probability distributions

Control	Implemented effect
Temperature 0.7 / 1.3	divide logits by temperature before softmax
Top-k 20	keep exactly the 20 highest-probability tokens, then renormalize
Top-p 0.9	keep the smallest descending-probability prefix whose cumulative mass reaches 0.9

Controlled autoregressive generation

Generation begins from the same held-out prompt derived from the first test context. Every configuration keeps the most recent 16 BPE tokens and adds exactly 40 new tokens.

Configuration	Method
greedy	argmax
temperature 0.7	seeded sampling after temperature scaling
top-k 20	seeded sampling from 20-token support
top-p 0.9	seeded nucleus sampling with p = 0.9

Attention inspection

Causal attention remains the same as Version 12: each block produces a 16 x 16 matrix, rows sum to 1, and all future positions are masked.

Included checks

- vocabulary and tensor shapes match the configuration
- the model remains two blocks, 23 variables, and 34,496 parameters
- LAST/BEST restoration and resume reach epoch 20
- perplexity equals exp(cross-entropy)
- top-5 accuracy is at least top-1 accuracy
- controlled distributions are finite, non-negative, and normalized
- top-k 20 keeps exactly 20 non-zero probabilities
- non-positive temperature is rejected
- seeded top-p sampling is reproducible
- all four generation modes add exactly 40 tokens
- causal attention remains normalized and future-masked
- the final test cell reports All checks passed

What Version 13 teaches

Version 13 separates two questions that were previously mixed together: how well the selected model predicts unseen tokens, and how generated text changes when the decoding rule changes. Training is deliberately kept fixed, so the new lesson is entirely about measurement and model use.

Version 13 base BEST

```
-> cross-entropy + perplexity
-> top-1 + top-5 token accuracy
-> temperature / top-k / top-p
-> reproducible sampling
```

Limitations

- only controlled WikiText-103 subsets are used
- the model remains deliberately tiny with a 16-token context
- token-level metrics do not directly measure factuality or usefulness
- temperature/top-k/top-p are simple decoding controls rather than advanced search methods
- sampling and greedy decoding can still produce repetitive text
- persistent checkpoints across separate Kaggle sessions remain out of scope

Conclusion

Version 13 extends the project from loss-only reporting to a clearer held-out evaluation and makes generation control explicit without changing the model or training budget. The separation between training, validation selection, test measurement, and decoding strategy is now visible in one compact experiment.

Future versions will introduce new components and gradually evolve the architecture.